

Uncertainty budgets for bubble wall velocities in cosmological phase transitions

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PLANCK2026 and 6th EuCAPT Symposium
CERN, 05/2026

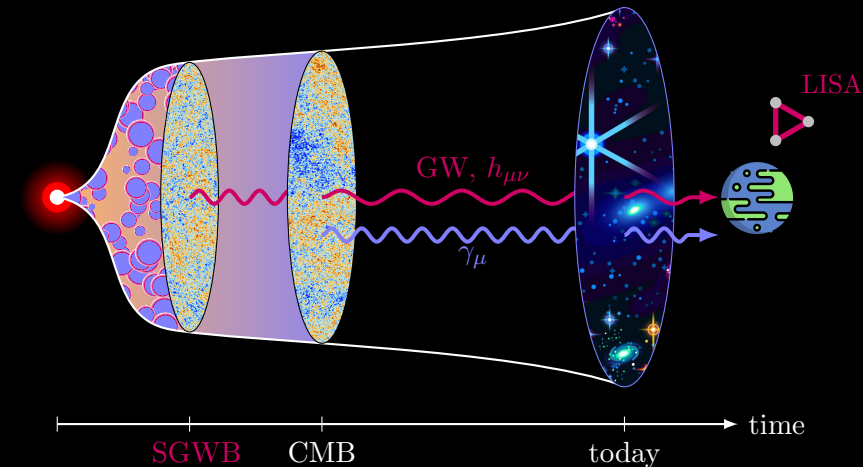


at weak coupling.

Supported by the SNSF under grant PZ00P2-215997.

Gravitational Waves (GWs) can be

sourced by early-universe phase transitions caused by new physics.



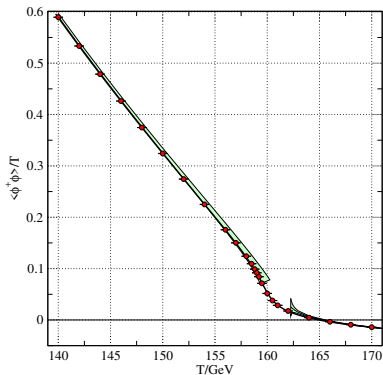
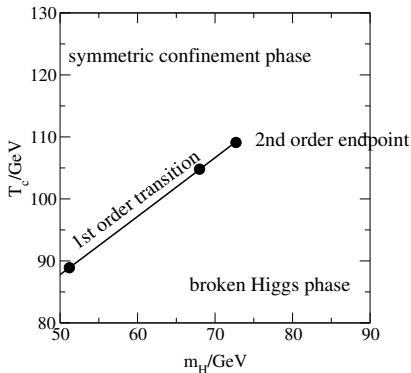
Cosmic Microwave Background (CMB)

Stochastic GW Background (SGWB)

Laser Interferometer Space Antenna (LISA)

GWs from a cosmological phase transition: New physics?

In Standard Model, EWSB occurs via a smooth crossover¹ but possible that it is first-order in Beyond the Standard Model (BSM) extensions.



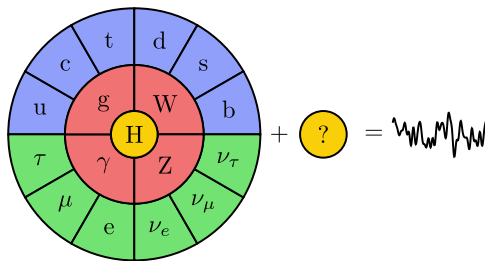
¹ Kajantie, Laine, Rummukainen, Shaposhnikov, *Is there a hot electroweak phase transition at $m_H \gtrsim m_W$?*, Phys. Rev. Lett. **77** (1996) 2887 [hep-ph/9605288], D'Onofrio, Rummukainen, *Standard model cross-over on the lattice*, Phys. Rev. D **93** (2016) 025003 [1508.07161]

GWs from a cosmological phase transition: New physics?

In Standard Model, EWSB occurs via a smooth crossover¹ but possible that it is first-order in Beyond the Standard Model (BSM) extensions.

Study **BSM physics** near EW scale in context of phase transitions:

- ▷ Light fields strongly coupled to Higgs
- ▷ Collider targets. BSM testing pipeline: Collider phenomenology



¹ Kajantie, Laine, Rummukainen, Shaposhnikov, *Is there a hot electroweak phase transition at $m_H \gtrsim m_W$?*, Phys. Rev. Lett. **77** (1996) 2887 [hep-ph/9605288], D'Onofrio, Rummukainen, *Standard model cross-over on the lattice*, Phys. Rev. D **93** (2016) 025003 [1508.07161]

Fingerprinting stochastic GW spectrum via few parameters

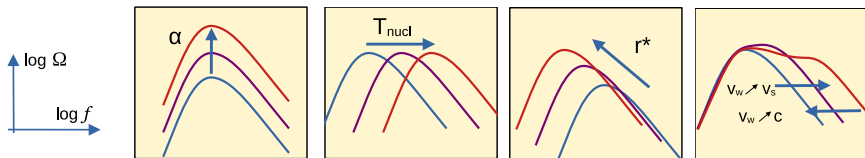
$T_\star$ reference temperature of the transition ($T_\star = T_p = T_{\text{percolation}}$),

α phase transition strength,²

β/H inverse duration of the transition (also $r_\star$),

c_s speed of sound in the plasma

$$h^2\Omega_{\text{GW}} = h^2\Omega_{\text{GW}}\left(T_\star, \alpha, \frac{\beta}{H}, c_s, \dots\right)$$



²**Caveat:** For supercooled transitions, α can be very large and fitted $h^2\Omega_{\text{GW}}$ have been tested for smaller α
figure by Kari Rummukainen

Fingerprinting stochastic GW spectrum via few parameters

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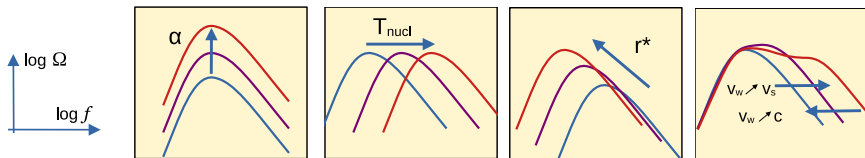
α phase transition strength,²

β/H inverse duration of the transition (also $r_\star$),

c_s speed of sound in the plasma

v_w terminal velocity of the bubble wall

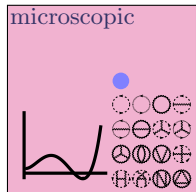
$$h^2\Omega_{\text{GW}} = h^2\Omega_{\text{GW}}\left(T_\star, \alpha, \frac{\beta}{H}, c_s, v_w, \dots\right)$$



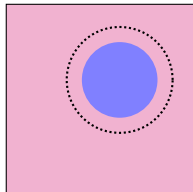
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figure by Kari Rummukainen

Connecting micro- and macrophysics

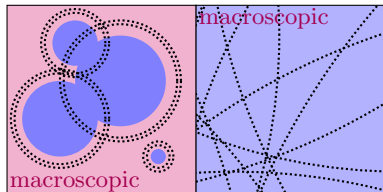
Large dynamic range to resolve Ω_{GW} at both low and high frequency.
Reliable GW predictions require control at **each** scale.



Bubble nucleation
thermodynamics



growth
wall-fluid system

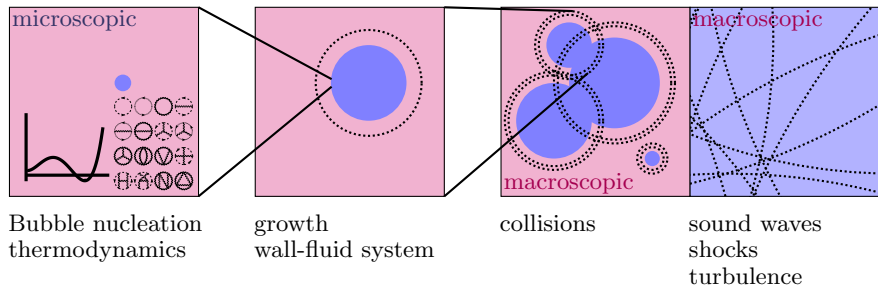


collisions

sound waves
shocks
turbulence

Connecting micro- and macrophysics

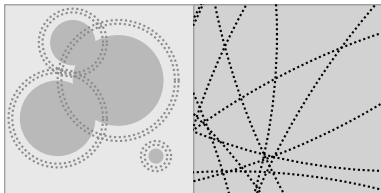
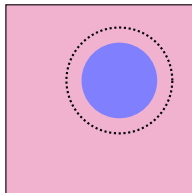
Large dynamic range to resolve Ω_{GW} at both low and high frequency.
Reliable GW predictions require control at **each** scale.



Strategy:

Effective field theory (EFT) to connect **micro-** and **macrophysics**.

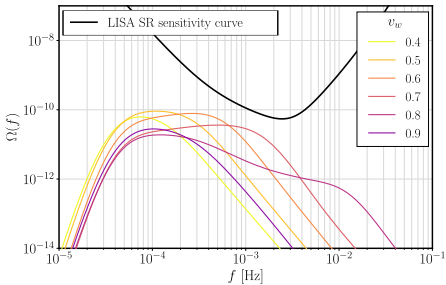
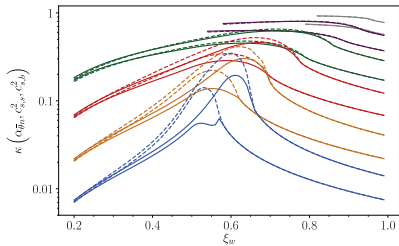
In between scales



Fingerprinting GW spectrum via bubble growth

For intermediate or weak strength transitions, velocity (v_w) of expanding bubbles determines the shape and strength of the GW signal.^{3,4}

In sound-shell model,⁵ v_w can well discriminate underlying models.



³ Giese, Konstantin, Schmitz, van de Vis, *Model-independent energy budget for LISA*, JCAP **01** (2021) 072 [2010.09744]

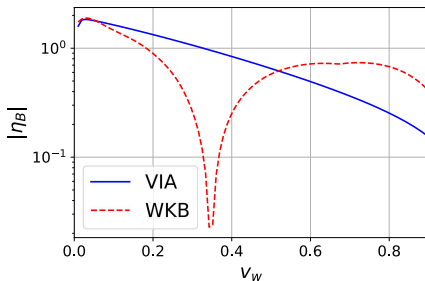
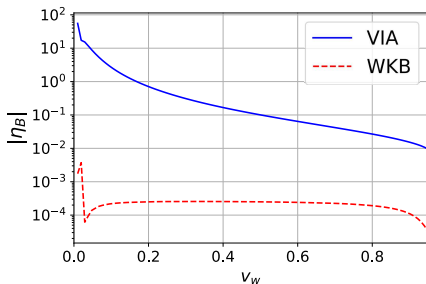
⁴ At fixed $\alpha = 0.2$, $r_* = 0.1$, $T_n = 100$ GeV from Gowling, Hindmarsh, *Observational prospects for phase transitions at LISA: Fisher matrix analysis*, JCAP **10** (2021) 039 [2106.05984]

⁵ Hindmarsh, Hijazi, *Gravitational waves from first order cosmological phase transitions in the Sound Shell Model*, JCAP **12** (2019) 062 [1909.10040]

Electroweak baryogenesis depends on the wall velocity⁶

Baryogenesis from CP violation via lepton higher-dimensional Yukawa.

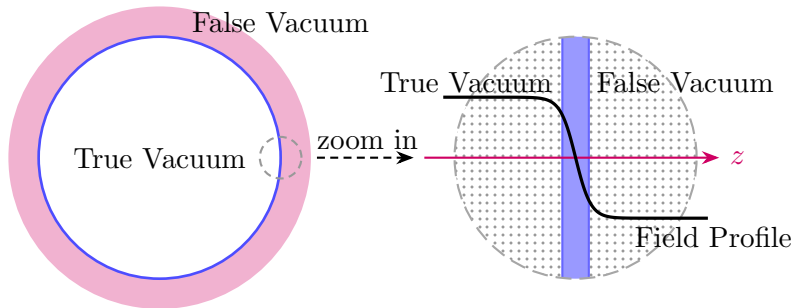
Baryogenesis by c - t or b - s flavor mixing.



⁶ Cline, Laurent, *Electroweak baryogenesis from light fermion sources: A critical study*, Phys. Rev. D **104** (2021) 083507 [2108.04249]; See talk by **M. R. Musolf** on **Thu 09:50**

Microscopic description of the bubble wall

The bubble wall is the region where the field(s) interpolate between the high- and low-temperature vacuum.



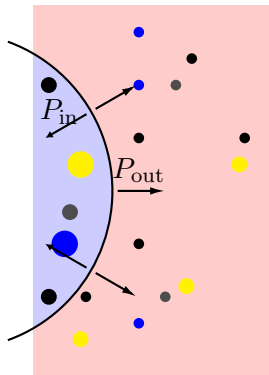
Bubbles typically reach terminal velocity before colliding if

$$|P_{\text{out}}| = |P_{\text{in}}|.$$

ΔP_{out} Vacuum energy release
(accelerating wall)

ΔP_{in} Particle friction on the wall
(decelerating wall)

ΔP_{in} Hydrodynamic backreaction
(decelerating wall)



Iteratively solve coupled **fluid**, **scalar field**, **Boltzmann** equations.⁷

Public software for v_w :⁸ **WA(GO)**.

⁷ Moore, Prokopec, *How fast can the wall move? A Study of the electroweak phase transition dynamics*, Phys. Rev. D **52** (1995) 7182 [hep-ph/9506475], Laurent, Cline, *First principles determination of bubble wall velocity*, Phys. Rev. D **106** (2022) 023501 [2204.13120],

⁸Schicho, Ekstedt, Gould, Hirvonen et al., *How fast does the WallGo? A package for computing wall velocities in first-order phase transitions*, JHEP **04** (2025) 101 [2411.04970].

Solve system of coupled equations

to compute v_w w/ **out-of-equilibrium contributions** at weak coupling.⁹

In scalar field EOM,¹⁰ $f = f_{\text{eq}} + \delta f$ and V_T contains f_{eq} :

$$\phi_{,\mu}^{\mu} + V_{,\phi}(T, \phi) + \sum_a \frac{dm_a^2}{d\phi} \int_{\mathbf{p}} \frac{1}{2E} \delta f^a(p^\mu, \xi) = 0 . \quad (1)$$

Particles in the plasma:

$$\frac{d}{dt} f = \hat{\mathbf{L}} f = \left[\partial_t + \dot{\mathbf{x}} \cdot \partial_{\mathbf{x}} + \dot{\mathbf{p}} \cdot \partial_{\mathbf{p}} \right] f = -\mathcal{C}[f] . \quad (2)$$

Temperature $T(\xi)$ and fluid $v(\xi)$ via energy-momentum conservation.

In practice determine v_w :

Vary wall parameters until all eqs. (1) + (2) + Hydro are satisfied.

⁹ Moore, Prokopec, *How fast can the wall move? A Study of the electroweak phase transition dynamics*, Phys. Rev. D **52** (1995) 7182 [hep-ph/9506475]

¹⁰Distance from wall $\xi = -\bar{u}_w^\mu x_\mu$ and $\bar{u}_w^\mu = \gamma_w(v_w, 0, 0, 1)$.

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Multiple out-of-equilibrium particles in the plasma:

$$\frac{d}{dt} f^a = \hat{\mathbf{L}} f^a = \left[\partial_t + \dot{\mathbf{x}} \cdot \partial_{\mathbf{x}} + \dot{\mathbf{p}} \cdot \partial_{\mathbf{p}} \right] f^a = -\mathcal{C}_a[\mathbf{f}] . \quad (2)$$

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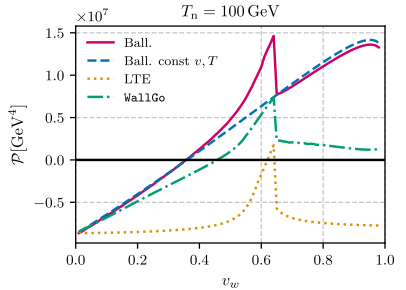
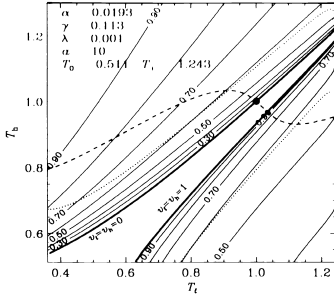
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Pinching v_w with LTE and ballistic limit

Upper bound. Large collision term ($\mathcal{C}_a \rightarrow \infty$ and $\delta f^a \rightarrow 0$):
Local thermal equilibrium (LTE).^{11,12}

Lower bound. Small collision term ($\mathcal{C}_a \rightarrow 0$):
Ballistic limit.¹³ Can solve Boltzmann equation analytically.



¹¹ Enqvist, Ignatius, Kajantie, Rummukainen, *Nucleation and bubble growth in a first order cosmological electroweak phase transition*, Phys. Rev. D **45** (1992) 3415

¹² Backreaction from hot fluid, and entropy production ($\Delta_{su} z \geq 0$) can still stop the wall c.f. Eriksson, Laine, *Entropy production at electroweak bubble walls from scalar field fluctuations*, JCAP **09** (2025) 027 [2507.07755]

¹³ Ai, Laurent, van de Vis, *Bounds on the bubble wall velocity*, JHEP **02** (2025) 119 [2411.13641] at $m_s = 120$ GeV and $\lambda_{hs} = \lambda_s = 1$

Scalar and Boltzmann equations source uncertainty in v_w

$$\phi_{,\mu}^{\mu} + V_{,\phi}(T, \phi) + \sum_a \frac{dm_a^2}{d\phi} \int_{\mathbf{p}} \frac{1}{2E} \delta f^a(p^\mu, \xi) = 0,$$

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Source of uncertainty	
Boltzmann	Linearization
Scalar EOM	

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Source of uncertainty	
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Source of uncertainty	
Boltzmann	Linearization Power enhanced collision integrals NLL collision integrals Particles in and out of equilibrium
Scalar EOM	

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Source of uncertainty	
Boltzmann	Linearization Power enhanced collision integrals NLL collision integrals Particles in and out of equilibrium Particle masses
Scalar EOM	

Scalar and Boltzmann equations source uncertainty in v_w

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Source of uncertainty	
Boltzmann	Linearization Power enhanced collision integrals NLL collision integrals Particles in and out of equilibrium Particle masses
Scalar EOM	Nucleation temperature (T_n)

Scalar and Boltzmann equations source uncertainty in v_w

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Source of uncertainty	
Boltzmann	Linearization Power enhanced collision integrals NLL collision integrals Particles in and out of equilibrium Particle masses
Scalar EOM	Nucleation temperature (T_n) Effective potential (V_{eff})

Scalar and Boltzmann equations source uncertainty in v_w

$$\phi_{,\mu}^{\prime\mu} + V_{,\phi}(T, \phi) + \sum_a \frac{dm_a^2}{d\phi} \int_{\mathbf{p}} \frac{1}{2E} \delta f^a(p^\mu, \xi) = 0,$$

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Source of uncertainty	
Boltzmann	Linearization
	Power enhanced collision integrals
	NLL collision integrals
	Particles in and out of equilibrium
Scalar EOM	Particle masses
	Nucleation temperature (T_n)
	Effective potential (V_{eff})
	Tanh-Ansatz

Scalar and Boltzmann equations source uncertainty in v_w

$$\phi_{,\mu}^{\mu} + V_{,\phi}(T, \phi) + \sum_a \frac{dm_a^2}{d\phi} \int_{\mathbf{p}} \frac{1}{2E} \delta f^a(p^\mu, \xi) = 0,$$

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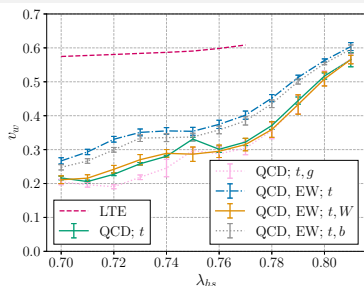
Source of uncertainty	
Boltzmann	Linearization
	Power enhanced collision integrals
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	Particles in and out of equilibrium
	Particle masses
Scalar EOM	Nucleation temperature (T_n)
	Effective potential (V_{eff})
	Tanh-Ansatz

Subset of out-of-equilibrium particles and interactions

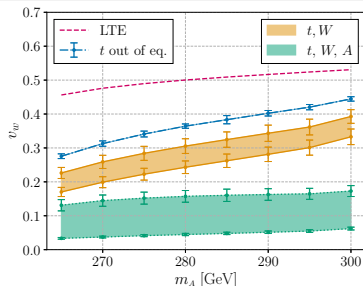
for largest masses and strongest couplings: **numerical cost vs. accuracy**.

Particles pushed out of equilibrium by passing bubble and non-constant fluid and temperature profiles, for xSM¹⁴ and IDM¹⁵.

xSM



IDM



¹⁴Singlet-extended SM: SM + Z_2 scalar singlet.

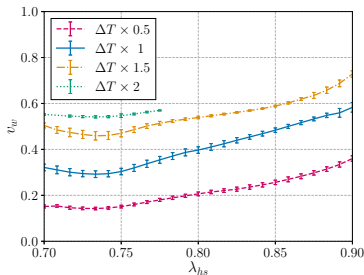
¹⁵Inert doublet model: a 2HDM with vev-less second doublet at $T = 0$.

Estimating higher-order thermodynamic corrections

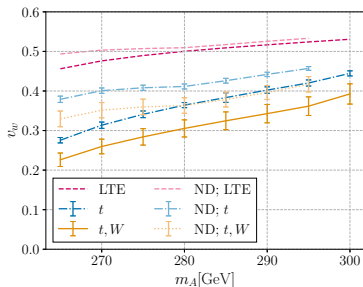
Varying the amount of supercooling in the xSM $T_n = \text{factor} \times T_n$.

Comparing V_{eff} with and without (ND) Daisy resummation in IDM.

xSM



IDM



Collision integrals are IR sensitive¹⁶

and affect the Boltzmann equation:

(P) *power enhanced*, naively quadratically IR divergent. Scaling:

$$\mathcal{C}_P/\delta f \sim g^4 T^4/m_{\text{eff}}^2 \sim g^2 T^2.$$

(LL) *leading-logarithmic enhanced*, naively logarithmically IR divergent. Scaling:

$$\mathcal{C}_{\text{LL}}/\delta f \sim g^4 T^2 \ln(T/m_{\text{eff}}) \sim g^4 T^2 \ln(1/g).$$

(NLL) *next-to-leading-logarithmic*, IR finite. Scaling:

$$\mathcal{C}_{\text{NLL}}/\delta f \sim g^4 T^2.$$

Regulate IR enhanced u - and t -propagators via HTL self-energy or asymptotic masses $m_{\text{eff}}^2 = \Pi_{\text{HTL}}(q_0 = \mathbf{q} \rightarrow \infty) \sim m_{\text{vac}}^2 + \mathcal{O}(g^2 T^2)$.

$$\left[\begin{array}{c} a \\ b \end{array} \rightarrow \begin{array}{c} c \\ d \end{array} \right]^2 \supset \frac{1}{t^2} \rightarrow \frac{1}{[t - \Pi_{\text{HTL}}(\boldsymbol{q}_0, \mathbf{q})]^2}.$$

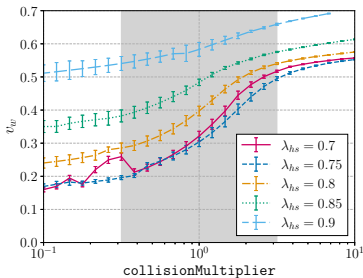
¹⁶ Arnold, Moore, Yaffe, *Transport coefficients in high temperature gauge theories. 1. Leading log results*, JHEP **11** (2000) 001 [hep-ph/0010177], Arnold, Moore, Yaffe, *Transport coefficients in high temperature gauge theories: (II) Beyond leading log*, JHEP **2003** (2003) 051 [0302165]

To estimate the error in v_w from LL collision terms

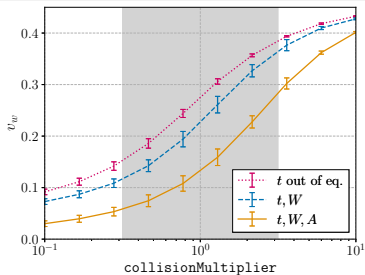
multiply all collision terms by a factor to mimic NLL result,

$$\mathcal{C}_{ab}[\delta f^b] \rightarrow \text{collisionMultiplier} \times \mathcal{C}_{ab}[\delta f^b].$$

xSM



IDM



Theoretical uncertainties in the wall velocity Δv_w ¹⁷

and the driving pressure ΔP , are quantified by

$$\Delta v_w \approx \Delta P \left(\frac{dP}{dv_w} \right)^{-1}, \quad \text{using } \text{WallGo}.$$

Source of uncertainty		$\Delta v_w / v_w$		
		xSM	IDM	$\mathcal{O}(\Delta P / P)$
Boltzmann	Linearization	0.1%	0.01%	$\mathcal{O}(\delta f_2 / f_{\text{eq}})$
	Power enhanced	90%	300%	$\mathcal{O}(\delta f_1 / f_{\text{eq}})$
	NLL collisions	—	20%	$\mathcal{O}(\delta f_1 / (f_{\text{eq}} \ln(1/g)))$
	Out-of-eq. particles	30%	400%	$\mathcal{O}(\delta f_1 / f_{\text{eq}})$
Scalar EOM	Nucleation (T_n)	100%	—	$\mathcal{O}(g)$
	Effective potential (V_{eff})	—	20%	$\mathcal{O}(g)$

¹⁷ van de Vis, Schicho, Niemi, Laurent, Hirvonen, Gould, *WallGo investigates: Theoretical uncertainties in the bubble wall velocity*, JHEP **04** (2026) 041 [2510.27691]

Theoretical uncertainties in the wall velocity Δv_w ¹⁷

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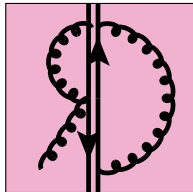
Improvement:

treat individual uncertainties or first-principle computations.¹⁸

¹⁷ van de Vis, Schicho, Niemi, Laurent, Hirvonen, Gould, *WallGo investigates: Theoretical uncertainties in the bubble wall velocity*, JHEP **04** (2026) 041 [2510.27691]

¹⁸ Ai, Carosi, Garbrecht, Tamarit, Vanvlasselaer, *Bubble wall dynamics from nonequilibrium quantum field theory*, JHEP **08** (2025) 077 [2504.13725], Ramsey-Musolf, Zhu, *Bubble wall velocity from Kadanoff-Baym equations: fluid dynamics and microscopic interactions*, [2504.13724]

Conclusions: Progress and future challenges



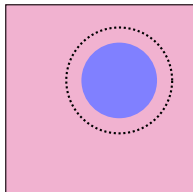
Precision thermodynamics of BSM theories via 3d EFT:

- ✱ higher dimensional operators,
- final perturbative order for nucleation rate,
- ↗ 3d EFT at NLO via DRalgo



Single bubble wall velocity v_w in the quasi-particle picture:

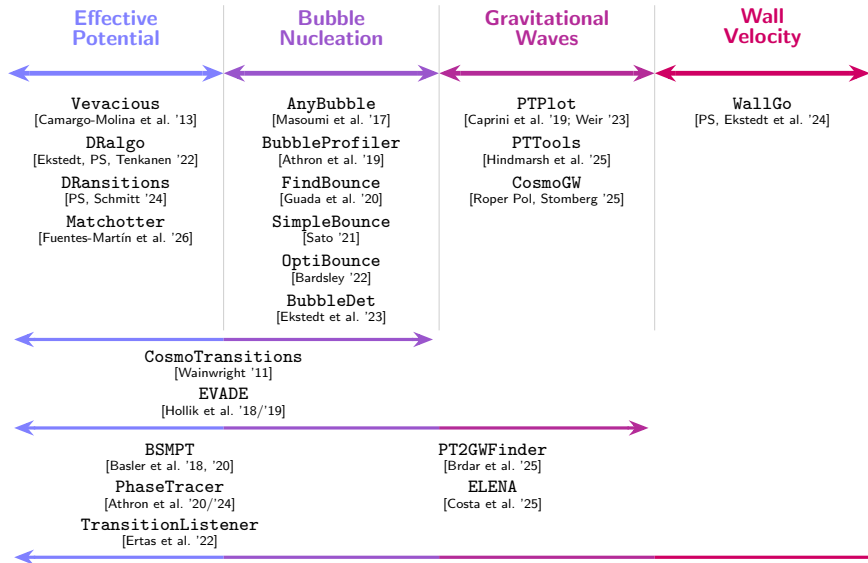
- ± uncertainty budget in phase transitions,
- ⚙ systematic framework developed in WallGo



Refining the velocity:

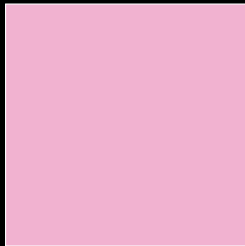
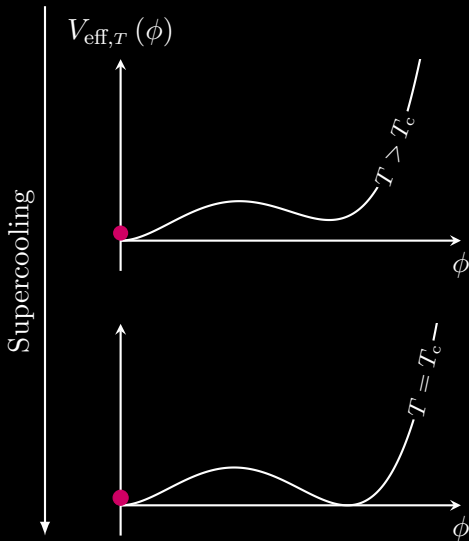
- ▷ First-principle computations of v_w via transport theory,
- ▷ inclusion of higher-order corrections.

One more package ...

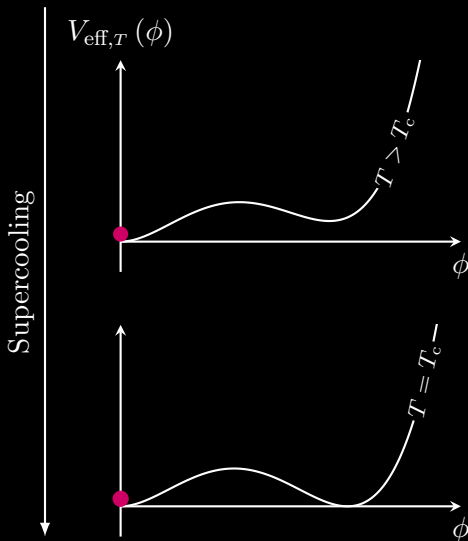


Backup

Cosmological first order phase transition

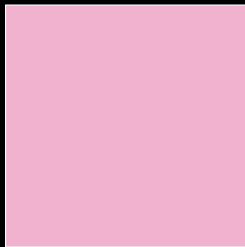


Cosmological first order phase transition

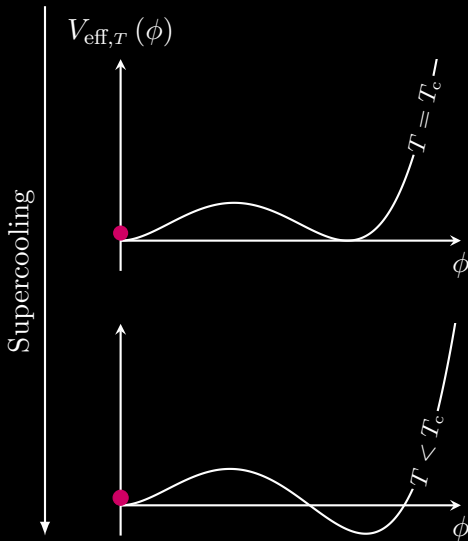


UV [thermal] corrections modify the zero- T potential and restore symmetry.

$$V_{\text{eff},T} \supset \text{[Feynman diagrams: tadpole, bubble, sunset, etc.]} \dots$$

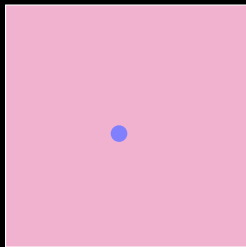


Cosmological first order phase transition

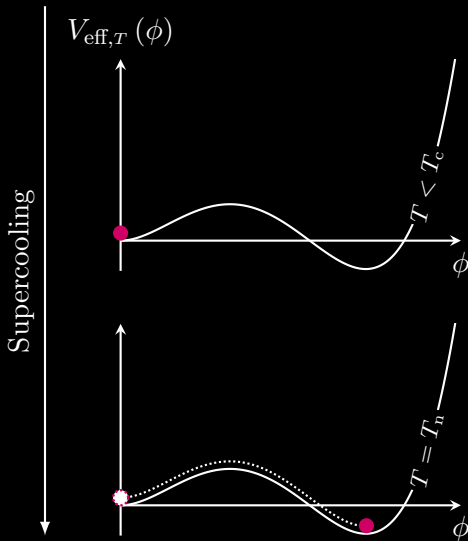


When the universe cools down, the field transitions.

If a barrier separates the phases, the transition is first order.

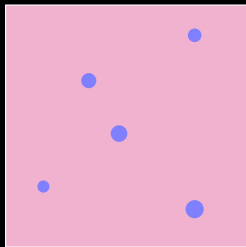


Cosmological first order phase transition



The phase transition proceeds via nucleation of bubbles.

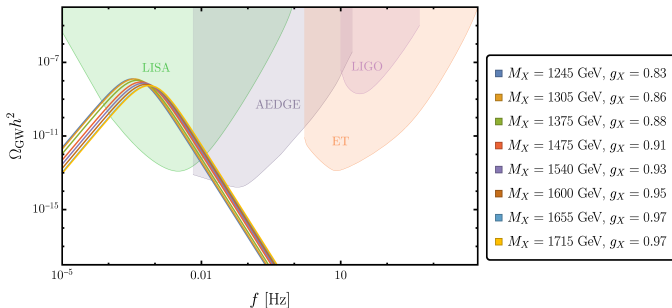
Part of the released energy can be converted into gravitational waves: $\square h_{ij} = 4\pi G T_{ij}$



The gravitational wave signal is stochastic

and the sum of independent events (interactions between the bubbles).
The spectral parameters contain information about the source.

Strong transitions with large signal in classically conformal models:¹⁹



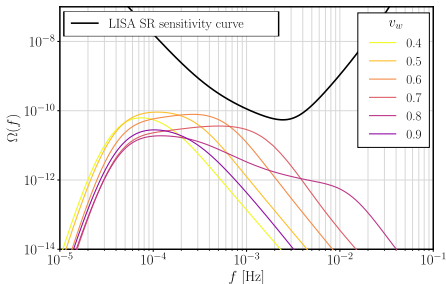
¹⁹ Kierkla, Karam, Swiezewska, *Conformal model for gravitational waves and dark matter: a status update*, JHEP **03** (2023) 007 [2210.07075], Kierkla, Schicho, Swiezewska, Tenkanen, van de Vis, *Finite-temperature bubble nucleation with shifting scale hierarchies*, JHEP **07** (2025) 153 [2503.13597], M_X being $SU(2)_X$ gauge boson mass.

The gravitational wave signal is stochastic

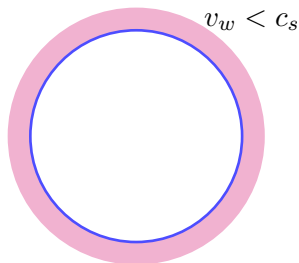
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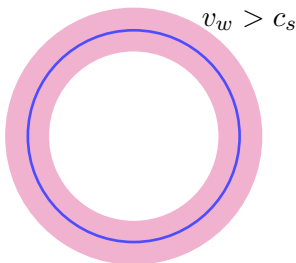
This talk: Intermediate-strength transitions



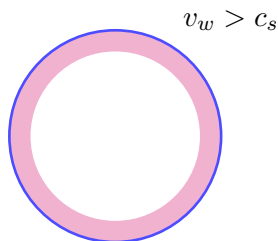
Bubble growth: relativistic combustion¹⁹



Deflagration



Supersonic deflagration
“Hybrid”

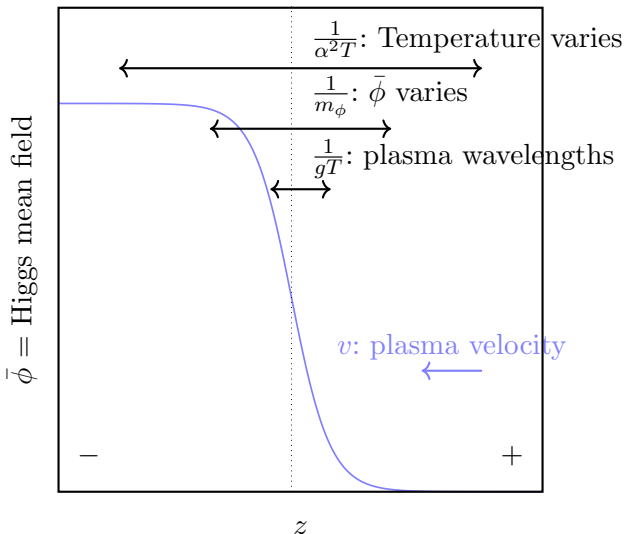


Detonation

¹⁹ Landau, Lifshitz, *Fluid mechanics*, Pergamon, 1959, Steinhardt, *Relativistic Detonation Waves and Bubble Growth in False Vacuum Decay*, Phys. Rev. D **25** (1982) 2074, Ignatius, Kajantie, Kurki-Suonio, Laine, *The growth of bubbles in cosmological phase transitions*, Phys. Rev. D **49** (1994) 3854 [astro-ph/9309059], Espinosa, Konstandin, No, Servant, *Energy Budget of Cosmological First-order Phase Transitions*, JCAP **06** (2010) 028 [1004.4187]

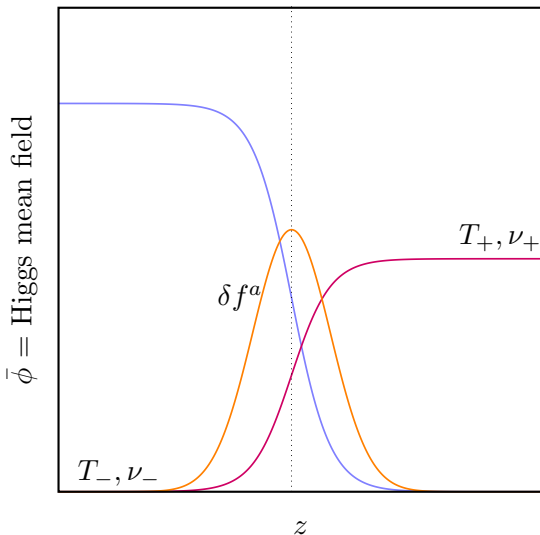
The relevant length scales during a phase transition

in the wall frame [plasma moves to the left; $\alpha = g^2/(4\pi)$].



The relevant length scales during a phase transition

in the wall frame [plasma moves to the left; $\alpha = g^2/(4\pi)$].



Effective description

with variables

$$\phi \text{ [Higgs]} , \quad T, u^\mu \text{ [plasma background]}$$

with leading-order matching coefficients

$$V(\phi, T) \text{ [potential]} , \quad \Upsilon(\phi, T) \text{ [friction]} .$$

Describe scalar field dynamics with covariant Langevin type EFT²⁰

$$\phi_{,\mu}^{\prime\mu} + V_{,\phi} + f(\Upsilon, \rho, \dots) = 0 ,$$

$T_{\mu\nu}^{\prime\mu} = 0$. The plasma reacts to the dynamics of ϕ

$$T_{\mu\nu}^\phi + T_{\mu\nu}^f = \phi_{,\mu}\phi_{,\nu} - g_{\mu\nu} \left(\frac{1}{2} \phi_{,\alpha}\phi^{,\alpha} - V(\phi) \right) + (e^f + p^f) u_\mu u_\nu - p^f g_{\mu\nu} .$$

Close to equilibrium one can use a quasi-particle description.

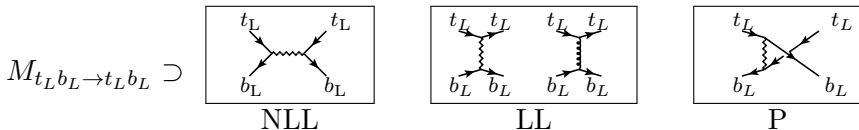
²⁰ Eriksson, Laine, *Entropy production at electroweak bubble walls from scalar field fluctuations*, JCAP **09** (2025) 027 [2507.07755]

IR sensitive contributions are softened by cancellations

in small momentum transfer regime of $ab \rightarrow cd$ scattering

$$\mathcal{C}_{ab \rightarrow cd}^{\text{lin}}[\delta f] \simeq \int_{\mathbf{p}_2, \mathbf{p}_3, \mathbf{p}_4} (2\pi)^4 \delta^4(p_1 + p_2 - p_3 - p_4) |M_{ab \rightarrow cd}|^2 \\ \times f_{\text{eq}}^a f_{\text{eq}}^b (1 \pm f_{\text{eq}}^c) (1 \pm f_{\text{eq}}^d) \left[\chi^a(\mathbf{p}_1) + \chi^b(\mathbf{p}_2) - \chi^c(\mathbf{p}_3) - \chi^d(\mathbf{p}_4) \right]$$

Keep only u - and t -channels and check if $\delta f^a \equiv \delta f^c$, $\delta f^b \equiv \delta f^d$, $\delta f^a \equiv \delta f^d$ to identify IR behavior of integral.



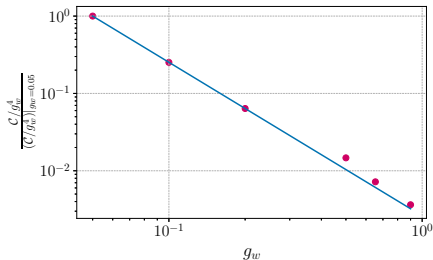
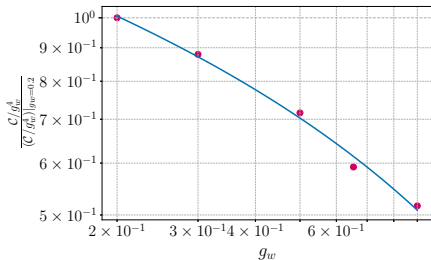
Not present in QCD since **particle species are conserved at vertices**.

IR sensitive contributions are softened by cancellations

in small momentum transfer regime of $ab \rightarrow cd$ scattering

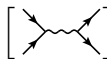
$$\mathcal{C}_{ab \rightarrow cd}^{\text{lin}}[\delta f] \simeq \int_{\mathbf{p}_2, \mathbf{p}_3, \mathbf{p}_4} (2\pi)^4 \delta^4(p_1 + p_2 - p_3 - p_4) |M_{ab \rightarrow cd}|^2 \\ \times f_{\text{eq}}^a f_{\text{eq}}^b (1 \pm f_{\text{eq}}^c) (1 \pm f_{\text{eq}}^d) \left[\chi^a(\mathbf{p}_1) + \chi^b(\mathbf{p}_2) - \chi^c(\mathbf{p}_3) - \chi^d(\mathbf{p}_4) \right]$$

Keep only u - and t -channels and check if $\delta f^a \equiv \delta f^c$, $\delta f^b \equiv \delta f^d$, $\delta f^a \equiv \delta f^d$ to identify IR behavior of integral.



Going beyond leading-logarithmic order

`WallGoMatrix` determines some next-to-leading-log (NLL) terms:

$$[\text{NLL}] \supset \left[\text{diagram} \right]^2 + [\text{LL}].$$


Dominant channels are $2 \rightarrow 2$ scatterings but $1 \rightarrow 2$ is NLL, $2 \rightarrow 3$ is coupling suppressed.

General momentum dependence in Mandelstam variables and $\Pi(p_0, \mathbf{p})$.

In `WallGoCollision 1.0.0` physical parameters must be $\sim T$. More scales that become relevant when opening more channels.

Example: vev-dependent masses in $1 \rightarrow 2$ scattering.

Affected transport coefficients²¹ by $\mathcal{O}(25\%)$.

No full NLL computation of the wall velocity has been done.

²¹ Arnold, Moore, Yaffe, *Transport coefficients in high temperature gauge theories: (II) Beyond leading log*, JHEP **2003** (2003) 051 [0302165]

